

From Cars to Clinics

Service-Oriented Middleware for Networked Medical Devices

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Research profile at Reutlingen University

Starting from my established work on safety-critical distributed systems, I see four accessible directions for applied research, student projects, and industry collaboration across the university.

Safe interoperability

- **Original challenge:** In a vehicle, independently developed software services must exchange sensor data safely and predictably in real time.
- **My experience:** Eight years of service-oriented, safety-critical middleware — SOME/IP, shared-memory communication, and sensor streams at Bosch.
- **Clinical direction:** Transfer these architectural principles to SDC/IEEE 11073-based device integration in connected clinical environments.

Digital twins

- **Original challenge:** How can we validate fleet software across thousands of